

MID TERM EXAMINATION

Subject: Linked Lists, Stacks, Queues

Paper: Mid Term Examination - Set A | Duration: 3 Hours | Max Marks: 38 | Date: 11-03-2026

Total Questions: 10 | Answer all questions.

Section 1: MCQ (4 Questions - 8 Marks)

Q1. [2 marks] (Easy)

In a singly linked list, which part of a node contains the address of the next node?

Options:

A) Data part B) Head pointer C) Next pointer D) Tail pointer

Q4. [2 marks] (Medium)

If you want to insert a new node before a specific node X in a singly linked list, what information is minimally required for an efficient $O(1)$ insertion operation (assuming X is not the head)?

Options:

A) Only a pointer to node X B) A pointer to the head of the list C) A pointer to the node preceding node X D) Pointers to both the head and tail of the list

Q6. [2 marks] (Medium)

Which of the following operations adds an element to the rear of a queue?

Options:

A) Pop B) Dequeue C) Enqueue D) Peek

Q8. [2 marks] (Hard)

Which of the following problems can be efficiently solved using a stack data structure?

Options:

A) Implementing a round-robin scheduler for processes B) Checking for balanced parentheses in an arithmetic expression C) Managing a print queue for documents D) Storing records for quick random access using an index

Section 2: Short Answer (6 Questions - 30 Marks)

Q2. [5 marks] (Easy)

Briefly explain the Last-In-First-Out (LIFO) principle as it applies to a Stack data structure.

Q3. [5 marks] (Medium)

Explain how a Circular Queue overcomes the "unutilized space" problem often encountered in a linear array-based queue when both front and rear pointers reach the end of the array.

Q5. [5 marks] (Hard)

Describe the algorithm to convert an infix expression to a postfix expression using a stack. Provide an example to illustrate the process.

Q7. [5 marks] (Medium)

Compare and contrast a Singly Linked List with a Doubly Linked List, highlighting one distinct advantage and one distinct disadvantage of Doubly Linked Lists over Singly Linked Lists.

Q9. [5 marks] (Easy)

Differentiate between an array-based and a linked-list-based implementation of a queue in terms of memory allocation and dynamic resizing capabilities.

Q10. [5 marks] (Hard)

Write pseudocode for an `insertInSortedList(head, newNode)` function that inserts `newNode` into a singly linked list such that the list remains sorted in ascending order. Assume `newNode` has a data and next field, and `head` points to the first node of the list (or NULL if empty).

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